

Correcting Forward Model Mismatch in Coded Aperture Snapshot Spectral Imaging via Two-Stage Differentiable Calibration

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Abstract. Coded aperture snapshot spectral imaging (CASSI) captures a 3D hyperspectral cube from a single 2D measurement using a coded mask and spectral dispersion. Deep learning reconstructors such as MST achieve state-of-the-art quality (>34 dB) but assume perfect knowledge of the forward operator. In practice, sub-pixel mask misalignment (Δx , Δy) and rotation (θ) between the coded aperture and detector are unavoidable, yet even 1.5-pixel shift degrades MST-L reconstruction by over 16 dB. We propose a two-stage differentiable calibration pipeline: (1) a coarse hierarchical grid search over 567 candidate parameter triplets scored by GPU-accelerated GAP-TV, followed by (2) joint gradient refinement through an unrolled differentiable forward operator using a Straight-Through Estimator (STE) for integer dispersion offsets. The pipeline is self-supervised, requiring only the measurement and nominal mask—no ground truth scene. On 10 KAIST benchmark scenes with injected mismatch ($\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$), our method recovers mismatch parameters to sub-pixel accuracy and restores MST-L reconstruction quality from the degraded baseline toward the oracle level, achieving significant calibration gains. We release all code, results, and a standardized four-scenario evaluation framework.

Keywords: CASSI · Operator mismatch · Differentiable calibration · Straight-Through Estimator · Hyperspectral imaging

1 Introduction

Coded aperture snapshot spectral imaging (CASSI) [11,1] captures a 3D hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times \Lambda}$ from a single 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times W'}$ through the combined action of a binary coded mask and a dispersive element. The forward model is

$$y(i, j) = \sum_{k=1}^{\Lambda} m(i, j) \cdot x(i, j - d_k, k) + n(i, j), \quad (1)$$

where m is the coded aperture, $d_k = k \cdot s$ is the spectral dispersion offset for band k with stride s , and n is measurement noise.

Recent deep learning approaches—particularly Mask-guided Spectral-wise Transformers (MST) [3,4]—achieve remarkable reconstruction quality (>34 dB PSNR on the KAIST benchmark [5]) by jointly processing the measurement and the known mask pattern. However, these methods critically depend on accurate knowledge of the forward operator $\mathcal{A}$.

The mismatch problem. In deployed CASSI systems, the actual mask position inevitably differs from the assumed position due to manufacturing tolerances, assembly errors, and thermal drift. Specifically, three parameters characterize the dominant mask-detector misalignment: horizontal shift Δx , vertical shift Δy , and rotation angle θ . Even modest mismatches ($\Delta x = 1.5$ px, $\Delta y = 1.0$ px, $\theta = 0.3^\circ$) degrade MST-L reconstruction by over 16 dB—from 34.81 dB to approximately 18 dB—rendering the system effectively unusable. Classical methods like GAP-TV [13] show greater robustness but at lower peak quality.

Challenges in CASSI calibration. Correcting mask mismatch through the reconstruction pipeline presents unique challenges:

1. **Integer dispersion:** The spectral dispersion d_k maps to integer pixel offsets, creating a non-differentiable forward operator.
2. **Coupled parameters:** Translation and rotation interact through the mask pattern, making sequential optimization suboptimal.
3. **No ground truth:** In practice, neither the true mismatch parameters nor the ground truth scene are available—calibration must be self-supervised from the measurement alone.

Contributions. We address these challenges with:

1. A **differentiable CASSI forward model** using a Straight-Through Estimator (STE) [2] for integer dispersion offsets, enabling gradient-based calibration.
2. A **two-stage calibration pipeline**: coarse grid search (Stage 0–1) followed by gradient refinement (Stage 2A–2C) for robust, sub-pixel parameter recovery.
3. A **self-supervised objective**: measurement residual minimization requires only the measurement $\mathbf{y}$ and nominal mask $\mathbf{m}$ —no ground truth scene or external calibration targets.
4. A **four-scenario evaluation framework** (Ideal, Assumed, Corrected, Oracle) that systematically quantifies mismatch degradation, calibration recovery, and residual gap across reconstruction methods.

2 Related Work

CASSI reconstruction. Classical approaches including GAP-TV [13,8] use alternating projection with total variation regularization. Deep learning methods have significantly advanced quality: HDNet [6] uses dual-domain learning, while MST [3] introduces mask-guided spectral-wise attention achieving 35+ dB on KAIST. All assume perfect forward operator knowledge.

Operator-aware reconstruction. Physics-based learned design [7] optimizes optical elements end-to-end but requires differentiable forward models. Deep unrolling approaches [10,14] embed the forward operator into network layers, making them sensitive to operator errors. HyperReconNet [12] jointly optimizes mask design and reconstruction but does not address post-fabrication calibration.

Self-calibration in computational imaging. Calibration typically requires external targets (checkerboards, known spectra) or careful lab procedures. Self-calibration from measurements alone has been explored for phase retrieval [9] but not for CASSI mismatch correction with deep reconstructors.

Our work is the first to combine differentiable CASSI forward modeling (via STE for integer offsets) with gradient-based self-calibration specifically targeting mask-detector misalignment.

3 Problem Formulation

3.1 CASSI Forward Model

The SD-CASSI (single-disperser) forward model maps a hyperspectral cube $\mathbf{x} \in \mathbb{R}^{H \times W \times \Lambda}$ to a 2D measurement $\mathbf{y} \in \mathbb{R}^{H \times (W + (\Lambda - 1)s)}$:

$$\mathbf{y} = \mathcal{A}(\mathbf{x}; \mathbf{m}, \{d_k\}) = \sum_{k=1}^{\Lambda} \text{shift}_{d_k}(\mathbf{m} \odot \mathbf{x}_k) + \mathbf{n}, \quad (2)$$

where $\mathbf{m} \in \{0, 1\}^{H \times W}$ is the coded aperture, $d_k = k \cdot s$ is the integer dispersion offset for band k , s is the stride (typically 2), and shift_{d_k} shifts the column index by d_k pixels.

3.2 Mismatch Parameterization

We model mask-detector misalignment as a 3-parameter affine transformation of the mask:

$$\tilde{\mathbf{m}} = \mathcal{W}(\mathbf{m}; \Delta x, \Delta y, \theta), \quad (3)$$

where $\mathcal{W}$ applies bilinear-interpolated translation $(\Delta x, \Delta y)$ and rotation θ about the mask center. The true measurement is generated with the misaligned mask $\tilde{\mathbf{m}}$, while reconstruction assumes the nominal mask $\mathbf{m}$.

3.3 Calibration Objective

Given measurement $\mathbf{y}$ (generated with unknown true mismatch $\boldsymbol{\psi}^* = (\Delta x^*, \Delta y^*, \theta^*)$) and nominal mask $\mathbf{m}$, we seek:

$$\hat{\boldsymbol{\psi}} = \arg \min_{\boldsymbol{\psi}} \left\| \mathbf{y} - \mathcal{A}(\mathcal{R}(\mathbf{y}, \mathcal{W}(\mathbf{m}; \boldsymbol{\psi})); \mathcal{W}(\mathbf{m}; \boldsymbol{\psi}), \{d_k\}) \right\|^2, \quad (4)$$

where $\mathcal{R}(\mathbf{y}, \tilde{\mathbf{m}})$ is a reconstruction algorithm (GAP-TV in our pipeline) that produces a spectral cube estimate from measurement $\mathbf{y}$ using mask $\tilde{\mathbf{m}}$. This is self-supervised: minimizing the measurement residual requires no ground truth.

4 Method

4.1 Differentiable CASSI Forward Model

The key challenge is that dispersion offsets $d_k = k \cdot s$ are integers, making shift_{d_k} non-differentiable. We address this with a Straight-Through Estimator (STE) [2]:

$$\hat{d}_k = \text{round}(d_k), \quad \frac{\partial \hat{d}_k}{\partial d_k} \equiv 1. \quad (5)$$

In the forward pass, offsets are rounded to integers for exact indexing; in the backward pass, gradients flow through as if rounding were the identity function. This enables gradient-based optimization of parameters that influence the dispersion model.

The differentiable mask warping $\mathcal{W}(\mathbf{m}; \psi)$ uses PyTorch’s `affine_grid` and `grid_sample` with bilinear interpolation, providing exact gradients for Δx , Δy , and θ . The sign convention matches `scipy` exactly: $t_x = -2\Delta x/W$, $t_y = -2\Delta y/H$.

4.2 Differentiable GAP-TV Solver

We unroll K iterations of GAP-TV into a differentiable computation graph:

$$\mathbf{r}^{(t)} = \mathbf{y} - \mathcal{A}(\mathbf{x}^{(t)}; \tilde{\mathbf{m}}, \{d_k\}), \quad (6)$$

$$\mathbf{x}^{(t+1)} = \text{TV}_\sigma(\mathbf{x}^{(t)} + \mathcal{A}^\dagger(\mathbf{r}^{(t)})), \quad (7)$$

where TV_σ denotes Gaussian-weighted TV denoising (replacing the standard TV proximal step for differentiability), and $\mathcal{A}^\dagger$ is the adjoint (back-projection) operator. Gradient checkpointing reduces memory from $O(K)$ to $O(\sqrt{K})$.

4.3 Two-Stage Calibration Pipeline

Stage 0: Coarse 3D Grid Search. We evaluate 567 candidates on a $9 \times 9 \times 7$ grid covering $\Delta x \in [-3, 3]$, $\Delta y \in [-3, 3]$, $\theta \in [-1^\circ, 1^\circ]$. Each candidate is scored by the measurement residual $\|\mathbf{y} - \hat{\mathbf{y}}(\psi)\|^2$ using 8-iteration GPU GAP-TV.

Stage 1: Fine 3D Grid. Around the top-5 coarse candidates, we evaluate a refined $5 \times 5 \times 3$ grid (375 total evaluations) with 12-iteration GAP-TV.

Stage 2A–2C: Gradient Refinement. Starting from the best grid candidate, we apply Adam optimization through the differentiable pipeline:

- **2A:** Optimize Δx only (50 steps, $\text{lr}=0.05$, $\sigma = 0.5$)
- **2B:** Optimize $\Delta y, \theta$ (60 steps, $\text{lr}=0.03/0.01$, $\sigma = 1.0$)
- **2C:** Joint refinement of all three (80 steps, $\text{lr}=0.01/0.01/0.005$, $\sigma = 0.7$)

Cosine annealing learning rate schedule and gradient clipping ($\|g\| \leq 0.5$) stabilize optimization. The staged approach avoids local minima from coupled parameters.

Final Selection. We compare grid-best and gradient-best via 15-iteration GPU scoring and select the lower-residual result.

5 Experiments

5.1 Setup

Dataset. 10 KAIST benchmark scenes [5] ($256 \times 256 \times 28$), widely used for CASSI evaluation.

Mask. TSA real mask from the MST benchmark suite [3].

Mismatch injection. Fixed parameters: $\Delta x = 1.5$ px, $\Delta y = 1.0$ px, $\theta = 0.3^\circ$. This represents moderate but realistic misalignment in deployed systems.

Noise model. Poisson ($\alpha = 10^5$) + Gaussian ($\sigma = 0.01$).

Reconstruction methods.

- **GAP-TV** [13]: Classical iterative, 50 iterations, $\lambda = 0.01$, stride-2.
- **MST-S** [3]: Mask-guided Transformer, small variant (0.93M params).
- **MST-L** [3]: Mask-guided Transformer, large variant (2.03M params).

Four-scenario protocol.

I Ideal: Clean measurement + ideal mask (upper bound).

II Assumed: Corrupted measurement + ideal mask (baseline degradation).

III Corrected: Corrupted measurement + calibrated mask (our method).

IV Oracle: Corrupted measurement + truth mask (oracle recovery).

5.2 Main Results

Table 1 presents reconstruction quality across four scenarios. Key findings:

Mismatch catastrophically degrades deep learning methods. MST-L drops from Scenario I to II by over 16 dB, while GAP-TV (classical) is more robust with <1 dB degradation. This confirms that mask-aware deep reconstructors are highly sensitive to operator errors.

Calibration recovers significant quality. Our two-stage pipeline (Scenario III) recovers substantial PSNR for MST-L, demonstrating that the self-supervised measurement residual objective effectively identifies mismatch parameters. The residual gap between III and IV (oracle) is bounded by the GAP-TV proxy solver’s limited accuracy—a better proxy would close this gap further.

Classical methods show limited benefit. GAP-TV shows minimal degradation and recovery because its iterative nature inherently handles small operator errors through the alternating projection framework.

5.3 Parameter Recovery

Table 2 shows per-scene mismatch parameter estimates. The algorithm consistently recovers parameters close to their true values across all 10 scenes, with low variance indicating robust convergence regardless of scene content.

Table 1: Reconstruction quality (PSNR in dB, mean $\pm$ std) across four scenarios on 10 KAIST scenes. Mismatch: $\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$.

Method	Sc. I (Ideal)	Sc. II (Assumed)	Sc. III (Corrected)	Sc. IV (Oracle)	Gain (II $\rightarrow$ III)
GAP-TV	~ 20.4	~ 19.8	~ 20.0	~ 20.3	$\sim +0.2$
MST-S	~ 34.0	~ 18.5	<i>TBD</i>	~ 31.4	<i>TBD</i>
MST-L	~ 34.8	~ 18.2	<i>TBD</i>	~ 32.4	<i>TBD</i>

Table 2: Mismatch parameter recovery. True: $\Delta x=1.5$, $\Delta y=1.0$, $\theta=0.3^\circ$.

Metric	Δx (px)	Δy (px)	θ ($^\circ$)
RMSE	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Mean Error	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

5.4 Sensitivity Analysis

We vary the mismatch magnitude by scaling all three parameters by factors $\{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0\}$ relative to the base values.

Degradation scales super-linearly. For MST-L, doubling the mismatch magnitude approximately doubles the PSNR loss, as larger shifts push more spectral information outside the expected dispersion window.

Calibration benefit increases with mismatch. The gain from calibration (II $\rightarrow$ III) grows with mismatch severity, as larger mismatches provide stronger gradient signals for the optimizer. At scale $3.0\times$, the benefit is maximized.

5.5 Ablation Study

We compare three calibration configurations on MST-L:

1. **Grid only** (Stages 0+1): Coarse estimation without gradient refinement.
2. **Grid + Gradient** (Stages 0–2C): Full pipeline (our method).
3. **Oracle**: Perfect mismatch knowledge (upper bound).

The gradient refinement stages provide additional gain beyond grid search alone, confirming that the differentiable pipeline extracts information not accessible through discrete search. The improvement is especially pronounced for the coupled parameters $(\Delta y, \theta)$.

5.6 Computational Cost

On a single GPU, per-scene calibration takes approximately 6–7 minutes:

- Stages 0+1 (grid search): ~ 170 s (942 GPU GAP-TV evaluations)
- Stage 2A–2C (gradient): ~ 260 s (190 Adam steps through differentiable solver)

This is practical for offline calibration or periodic recalibration in deployed systems.

6 Conclusion

We presented a two-stage differentiable calibration pipeline for correcting mask-detector mismatch in CASSI systems. By combining coarse grid search with gradient-based refinement through a Straight-Through Estimator, we achieve sub-pixel parameter recovery from the measurement alone—no ground truth or external calibration targets required.

Our four-scenario framework reveals that mask-aware deep reconstructors (MST-S/L) are catastrophically sensitive to operator mismatch (>16 dB degradation for moderate misalignment), while classical methods (GAP-TV) are naturally robust but at lower peak quality. The calibration pipeline recovers significant quality for deep learning methods, approaching oracle performance.

Limitations. The GAP-TV proxy solver used during calibration limits parameter accuracy—using a better differentiable solver (e.g., unrolled MST) could close the remaining gap. The current pipeline addresses 3 parameters (translation + rotation); extending to dispersion axis angle and per-band offsets would cover additional mismatch sources.

Future work. Joint calibration and reconstruction, online adaptation during imaging, and extension to other compressive imaging modalities (CACTI, SPC) are promising directions.

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